CS228 Logic for Computer Science
Problem Sheet 10

Conventions.

- Notation is as in Lectures 23–24. A transition system (model) is $M = (S, \to, L)$, where S is a set of states, $\to \subseteq S \times S$ is a transition relation such that every state has a successor, and $L : S \to \mathcal{P}(AP)$ labels every state with the atomic propositions that hold there. We also write TS for a transition system.
- A path is $\pi = s_1 \to s_2 \to \cdots$ and π^i is its suffix starting at s_i. LTL formulas are evaluated on paths ($\pi \models \varphi$); $M, s \models \varphi$ means that every path of M starting from s satisfies φ. CTL formulas are evaluated at states ($M, s \models \varphi$).
- We use the temporal operators X (Next), F (Eventually), G (Always), U (Until), W (Weak Until) and R (Release), where $\varphi\, W\, \psi := \varphi\, U\, \psi \vee G\, \varphi$ and $\varphi\, R\, \psi := \neg(\neg\varphi\, U\, \neg\psi)$.
- Brackets may be dropped using the binding rules: unary connectives ($\neg$, X, F, G in LTL; $\neg$, AX, EX, AF, EF, AG, EG in CTL) bind most tightly, then U, then $\vee, \wedge$, then $\to$.

Part 1

1. A printer is idle until a job arrives. The job is queued, and then the printer starts printing it. When the printing is done, the printer is idle again, but a paper jam can occur while printing. A jammed printer stays jammed until the jam is cleared, after which it is idle again. We model this as a transition system M over $AP = \{\mathbf{ready}, \mathbf{busy}, \mathbf{error}\}$. Figure 1 shows each state with its name and, in blue, its label. The event names on the edges are only comments: the transitions of M are just pairs of states.

   (a) List all elements of $\mathcal{P}(AP)$.

   (b) Write the formal description of M, i.e., S, $\to$ and L.

   (c) Which elements of $\mathcal{P}(AP)$ are the label of some state of M, and which are not?

2. For each of the following formulas over $AP = \{p, q, r\}$, decide whether it is a well-formed LTL formula and whether it is a well-formed CTL formula. For every formula that is well-formed (in LTL, in CTL, or in both), draw its parse tree and list all its subformulas. For every logic in which a formula is not well-formed, say what fails.

   (a) $\neg(p \vee q) \to r$

   (b) $G\,(p \to F\,(q\ U\ r))$

   (c) $A[p\ U\ E[q\ U\ r]]$

   (d) $AG\,F\,p$

   (e) $A[(r\ U\ q) \wedge (p\ U\ r)]$

   (f) $F\,p \to G\,r \vee \neg q\ U\ p$

3. Consider again the printer transition system M of Problem 1 (Figure 1), and the following LTL formulas over AP.

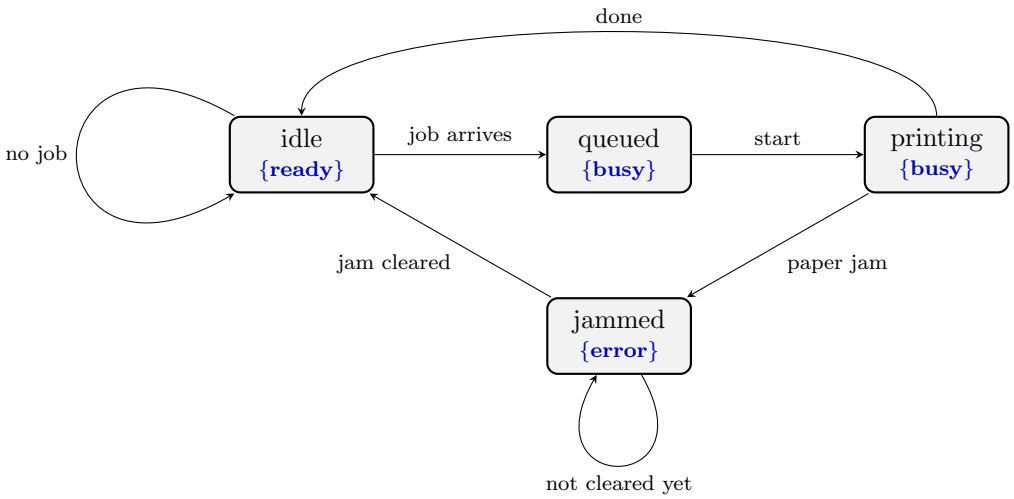


Figure 1: The printer transition system M.

$$\varphi_1 := X\,(\mathbf{ready} \vee \mathbf{busy})$$
$$\varphi_2 := G\,(\mathbf{error} \rightarrow X\,\mathbf{error})$$
$$\varphi_3 := F\,G\,\mathbf{ready}$$
$$\varphi_4 := (\mathbf{ready} \vee \mathbf{busy})\ U\ \mathbf{error}$$
$$\varphi_5 := G\,F\,\mathbf{busy} \rightarrow G\,F\,\mathbf{ready}$$

(a) Say in one line of English what each of the formulas $\varphi_1, \ldots, \varphi_5$ expresses.

(b) Consider the following path of M (the printer stays idle from the sixth state on):

$$\pi = \mathit{idle} \rightarrow \mathit{queued} \rightarrow \mathit{printing} \rightarrow \mathit{jammed} \rightarrow \mathit{jammed}$$
$$\rightarrow \mathit{idle} \rightarrow \mathit{idle} \rightarrow \mathit{idle} \rightarrow \cdots$$

Which of the formulas $\varphi_1, \ldots, \varphi_5$ does π satisfy? Justify each answer.

(c) Which of the formulas $\varphi_1, \ldots, \varphi_5$ hold in M at the state *idle*? (Recall that this means that the formula holds on every path of M starting at *idle*.) Justify each answer; if one fails, give a path of M on which it fails. Compare your answers with those in (b).

(d) Do the formulas $\varphi_1, \ldots, \varphi_5$ also hold in M if we start at a different state: at *queued*, and at *printing*? Justify your answers, and say which verdicts differ from those at *idle*.

(e) Write an LTL formula over AP for each of the following properties, and check whether it holds in M at the state *idle*. Justify your answer; if it fails, give a path of M on which it fails.

(i) The printer is ready infinitely often.

(ii) Whenever the printer is in error, it is eventually ready again.

4. Consider again the printer transition system M of Problem 1 (Figure 1).

(a) For each of the following CTL formulas, say in one line of English what it expresses, and decide whether it holds in M at the state *idle*. Justify each answer; if one fails, say why.

(i) $EX\,\mathbf{busy}$

(ii) $AF\,\mathbf{busy}$

(iii) $EG\,\neg\mathbf{error}$

(iv) $E[(\mathbf{ready} \vee \mathbf{busy})\ U\ \mathbf{error}]$

(v) $AG\,(\mathbf{busy} \rightarrow EF\,\mathbf{ready})$

(vi) $AG\,(\mathbf{busy} \rightarrow A[\mathbf{busy}\ U\ (\mathbf{ready} \vee \mathbf{error})])$

(b) Write a CTL formula over AP for each of the following properties, and check whether it holds in M at the state *idle*. Justify your answer; if it fails, say why.

(i) From every state, it is possible for the printer to become ready.

(ii) Whenever the printer is busy, then on every possible continuation it eventually becomes ready or goes into error.

5. Recall that LTL formulas φ and ψ are equivalent, $\varphi \equiv \psi$, if for all models M and all paths π in M we have $\pi \models \varphi$ iff $\pi \models \psi$. Which of the following LTL equivalences hold? Prove the ones that do, using the semantics of LTL as in Lecture 24. For the ones that do not, exhibit a model M and a path π in M on which the two sides differ.

(a) $X\,(\varphi\ U\ \psi) \equiv (X\varphi)\ U\ (X\psi)$

(b) $\varphi\ U\ \psi \equiv \psi \vee (\varphi \wedge X\,(\varphi\ U\ \psi))$

(c) $G\,F\,(\varphi \wedge \psi) \equiv G\,F\,\varphi \wedge G\,F\,\psi$

6. Two CTL formulas φ, ψ are equivalent, $\varphi \equiv \psi$, if for all models M and all states s we have $M, s \models \varphi$ iff $M, s \models \psi$. We did not cover CTL equivalences in class, but the proofs use the same idea as for LTL: take an arbitrary model M and an arbitrary state s, and show, using the semantics of CTL (Lecture 24), that $M, s \models \varphi$ iff $M, s \models \psi$ (for example, by negating the quantifiers over paths and over the positions on a path). Prove each of the following CTL equivalences.

(a) $\neg AF\varphi \equiv EG\neg\varphi$

(b) $\neg EF\varphi \equiv AG\neg\varphi$

(c) $\neg AX\varphi \equiv EX\neg\varphi$

(d) $AF\varphi \equiv A[\top\ U\ \varphi]$

(e) $EF\varphi \equiv E[\top\ U\ \varphi]$

(f) $AF\varphi \equiv \varphi \vee AX\,AF\varphi$

Part 2

7. Consider an elevator system that services 4 floors numbered 0 through 3. There is an elevator door at each floor with a call-button and an indicator light that signals whether or not the elevator has been called. In each step the elevator can open the door at its floor, close it again, move to an adjacent floor, or stay as it is; it moves only while all doors are closed.

(a) Draw a transition system M that models the doors and the movement of this elevator (ignore the call-buttons and lights). Use the atomic propositions $\mathbf{at}_f$ (the elevator is at floor f) and $\mathbf{open}_f$ (the door at floor f is open), for $f = 0, 1, 2, 3$, and label every state. Moving to an adjacent floor takes one step.

(b) How many states would the transition system have if it also recorded which call-buttons have been pressed, i.e., which indicator lights are on? Count every combination of the elevator's floor, the position of its door, and the on/off setting of each light, and explain how you count.

(c) Describe the following properties of the elevator system as LTL formulae, using the propositions of (a) and, where needed, $\mathbf{req}_f$ (the call-button at floor f has been pressed and not yet served). Try to use the minimum number of propositions in your AP. (The model M of (a) has no call-button propositions, so the formulas that mention $\mathbf{req}_f$ can only be written, not checked on M.)

(i) The doors are "safe", i.e., a floor door is never open if the elevator is not present at the given floor.

(ii) A requested floor will be served sometime.

(iii) Again and again the elevator returns to floor 0.

(iv) When the top floor is requested, the elevator serves it immediately and does not stop on the way there.

(d) (LTL) An upwards moving elevator at floor 1 does not change its direction if someone has pressed the button for floor 3. Come up with the additional atomic propositions you need and write an LTL formula for this specification.

(e) (CTL) The elevator can remain idle at floor 2 with its doors closed. Come up with atomic propositions and write a CTL formula for this specification.

(f) Let s_0 be the state of M from (a) in which the elevator is at floor 0 with all doors closed. Does $G\,F\,\mathbf{at}_0$ ("again and again the elevator returns to floor 0") hold in M at s_0? Does $AG\,EF\,\mathbf{at}_3$ hold in M at s_0? Justify your answers.

8. Recall that a set of connectives is adequate for a logic if all the other connectives of the logic can be expressed using only the connectives in the set. Show that the following sets are adequate.

(a) $\{\top, \neg, \wedge, X, U\}$ for LTL.

(b) $\{\bot, \rightarrow, X, R\}$ for LTL, where R is the Release operator: $\varphi\ R\ \psi := \neg(\neg\varphi\ U\ \neg\psi)$.

(c) $\{EG, EU, EX\}$ for the temporal connectives of CTL (together with the propositional connectives). In particular, express AX, AF, EF, AG and AU using only these. Are there yet other adequate sets?

9. In this problem we define new operators of LTL in terms of the standard ones X, F, G, U, W, R.

(a) 'Morally' there ought to be a dual for the Weak Until (W) operator. Work out what it might mean mathematically, express it in terms of standard LTL operators, and pick a symbol based on the first letter of its meaning.

(b) Let φ and ψ be LTL formulae. Consider the following new operators. Show that they are derived, i.e., give an LTL formula that expresses each of them.

(i) "At next" $\varphi\ N\ \psi$: at the next time where ψ holds, φ also holds.

(ii) "While" φ Wh ψ: φ holds at least as long as ψ does.

(iii) "Before" $\varphi\ B\ \psi$: if ψ holds sometime, φ does so before.

10. An LTL formula ψ and a CTL formula φ are equivalent, $\psi \equiv \varphi$, if for all models M and all states s we have $M, s \models \psi$ iff $M, s \models \varphi$ (recall that for an LTL formula, $M, s \models \psi$ means that all paths of M from s satisfy ψ).

(a) Is $F\,G\,p \equiv AF\,AG\,p$? Exhibit a model that distinguishes them.

(b) Let $AP = \{\mathbf{enabled}, \mathbf{running}\}$. Write an LTL formula for the property: if a process is enabled infinitely often, then it runs infinitely often. What does the CTL formula $AG\,AF\,\mathbf{enabled} \rightarrow AG\,AF\,\mathbf{running}$ say? Show that it is not equivalent to your LTL formula.